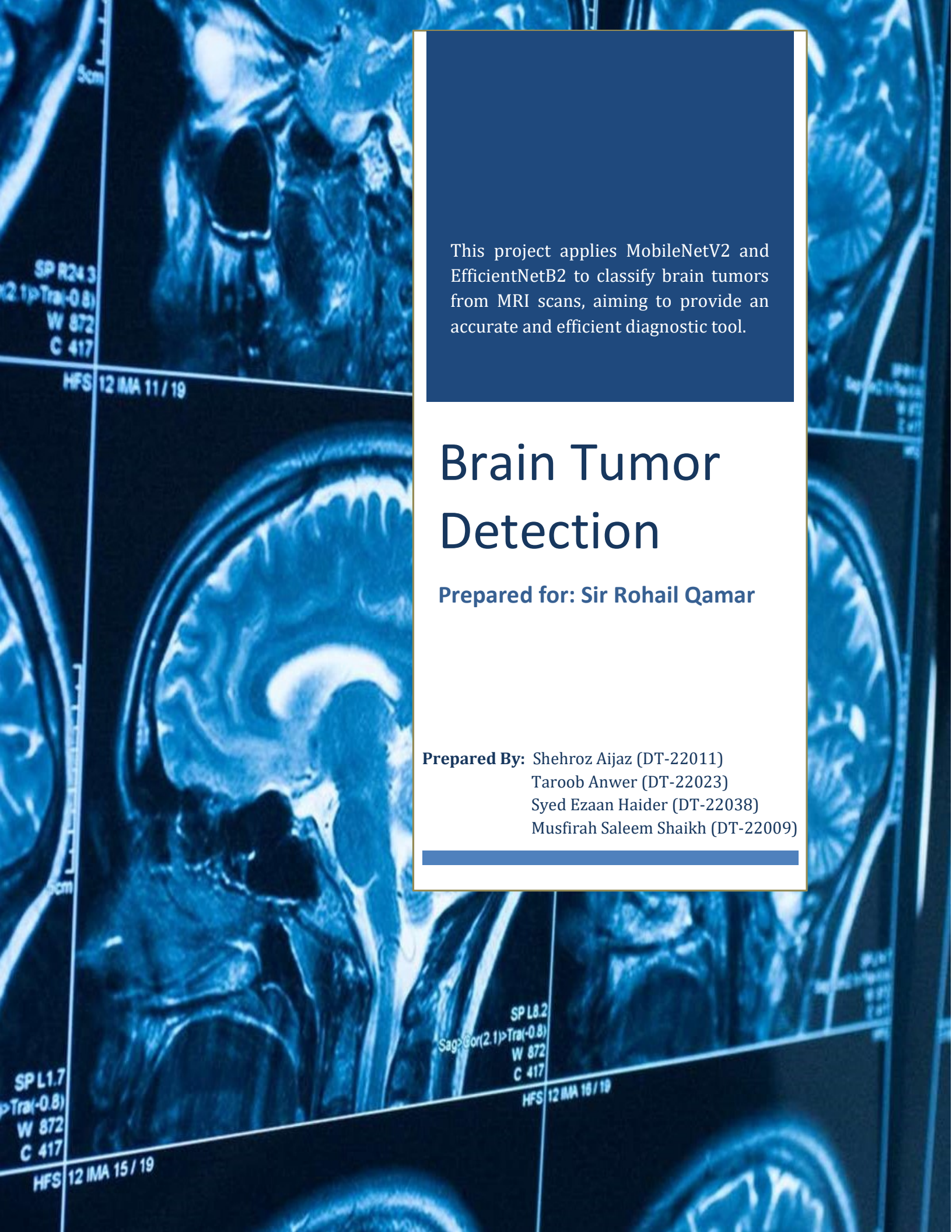


The background of the slide is composed of several overlapping MRI brain scan images. These include axial, sagittal, and coronal views of the brain, showing various internal structures like the ventricles and brain tissue. The images are in shades of blue and white, typical of medical MRI scans. Some text from the scan overlays is visible, such as 'SP R24.3', 'W 872', 'C 417', and 'HFS 12 IMA 11 / 19'.

This project applies MobileNetV2 and EfficientNetB2 to classify brain tumors from MRI scans, aiming to provide an accurate and efficient diagnostic tool.

Brain Tumor Detection

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Brain Tumor Detection using MobileNetV2 and EfficientNetB2

Abstract

This project explores the application of deep learning techniques, specifically transfer learning with MobileNetV2 and EfficientNetB2, to detect brain tumors from MRI images. The classification is performed among four categories: glioma, meningioma, pituitary tumors, and no tumor. By leveraging pre-trained models from ImageNet and augmenting the dataset for robustness, the model aims to achieve high accuracy and generalizability. Additional techniques such as fine-tuning, class balancing, and early stopping are integrated to ensure efficient and effective training. This report outlines the methodology, key features, results, and future recommendations.

1. Introduction

Brain tumors are abnormal growths within the brain, posing severe threats to health and quality of life. Early and accurate detection significantly improves patient outcomes, enabling prompt medical intervention. Magnetic Resonance Imaging (MRI) is a critical diagnostic tool widely employed for tumor detection, providing detailed visualization of soft tissue structures within the brain. However, manual interpretation of MRI scans by radiologists can be challenging, labor-intensive, and susceptible to diagnostic variability due to human error. To overcome these limitations, automated diagnostic systems utilizing artificial intelligence (AI), particularly deep learning methods, have become increasingly prevalent.

This project focuses on automated brain tumor detection employing two advanced convolutional neural network (CNN) architectures—MobileNetV2 and EfficientNetB2. These models have gained popularity due to their high accuracy and computational efficiency, making them suitable for practical clinical applications. Both architectures leverage transfer learning, a process in which pre-trained models previously trained on large-scale datasets like ImageNet are fine-tuned for specific medical imaging tasks. Transfer learning significantly reduces the need for extensive training data, which is beneficial since medical datasets are often limited in size.

To increase model accuracy and robustness, extensive data augmentation techniques, such as image rotation, flipping, shifting, zooming, and brightness adjustments, were utilized. This enhances the models' generalization capabilities, allowing accurate predictions even with variability in MRI data. Moreover, class imbalance, a common issue in medical datasets, was addressed through class weight balancing. Additionally,

various modern deep learning practices—including batch normalization, dropout regularization, early stopping, and L2 regularization—were integrated into the training pipeline.

Ultimately, this work aims to develop accurate, robust, and clinically applicable AI-driven diagnostic models for brain tumor detection, thereby supporting radiologists and potentially improving patient outcomes through timely and precise diagnosis.

2. Literature Review

Automated detection of brain tumors using deep learning and convolutional neural networks (CNNs) has attracted significant research interest due to its potential to assist clinicians by enhancing diagnostic accuracy and efficiency. Traditional diagnostic methods primarily depend on manual interpretation of MRI scans by radiologists, which is prone to errors, observer variability, and significant time investment (Litjens et al., 2017). Recently, the advent of CNN-based deep learning techniques, notably transfer learning, has substantially improved the accuracy and speed of medical image analysis tasks (Esteva et al., 2017).

CNNs have shown remarkable success in medical image classification tasks, such as detecting and classifying tumors, due to their ability to learn hierarchical features directly from raw imaging data. Studies such as those by Amin et al. (2020) and Sultan et al. (2019) have demonstrated that CNN-based approaches outperform traditional machine learning techniques (e.g., SVM and random forests) by significant margins, achieving higher sensitivity, specificity, and overall accuracy.

Transfer learning has emerged as a valuable method for medical imaging tasks, especially when datasets are limited. According to Tajbakhsh et al. (2016), employing transfer learning from large-scale datasets like ImageNet allows neural networks to transfer learned feature representations, drastically reducing the amount of training data needed and accelerating model convergence. MobileNetV2 and EfficientNet, in particular, have emerged as prominent choices for transfer learning due to their computational efficiency and high accuracy. MobileNetV2 utilizes depth-wise separable convolutions, making it suitable for deployment on resource-constrained platforms (Sandler et al., 2018). EfficientNet, introduced by Tan and Le (2019), employs compound scaling of model dimensions—depth, width, and resolution—achieving state-of-the-art performance on ImageNet classification tasks while maintaining computational efficiency.

Various data augmentation strategies have been successfully applied to enhance the generalization capabilities of CNN models. Techniques like rotation, zooming, flipping, and brightness adjustments are known to significantly improve robustness and performance, particularly in small and imbalanced medical imaging datasets (Shorten & Khoshgoftaar, 2019).

Overall, existing literature emphasizes that deep learning models, supported by transfer learning and data augmentation, provide robust solutions to challenges inherent in medical imaging and have great potential to improve diagnostic accuracy and clinical decision-making processes in brain tumor detection.

3. Methodology

3.1 Dataset

The dataset used in this research was sourced from the publicly available "Brain Tumor MRI Dataset" hosted on Kaggle. It consists of MRI images categorized into four distinct classes: glioma tumor, meningioma tumor, pituitary tumor, and no tumor. A total of 7,022 MRI scans were provided, which were balanced across these categories, ensuring equitable representation and minimizing potential biases during the model training process.

Before the images were used for training the deep learning models, a series of preprocessing steps were applied. Each image was resized to a standardized dimension of 224x224 pixels, a commonly accepted input size for deep convolutional neural networks. This resizing step ensures consistency and facilitates efficient computation. Subsequently, pixel values in the images were normalized to range between 0 and 1, enhancing numerical stability during training and leading to improved convergence rates. The processed images were then converted into NumPy arrays to ensure compatibility with the TensorFlow/Keras deep learning framework.

To robustly evaluate the performance and generalization capability of the models, the dataset was split into training, validation, and test subsets using stratified sampling. Specifically, 70% of the images were allocated to the training set, 15% to validation, and the remaining 15% to the test set. Stratified sampling was employed to maintain proportional class representation across each subset, ensuring that models learned evenly from each tumor category.

3.2 Model Architectures

Two sophisticated convolutional neural network architectures, MobileNetV2 and EfficientNetB2, were selected based on their demonstrated efficacy in image classification tasks, particularly within resource-constrained environments.

MobileNetV2 employs depthwise separable convolutions combined with linear bottleneck structures, significantly reducing computational requirements while preserving accuracy. This makes it highly suitable for real-time applications, including deployment on mobile or embedded devices, where computational resources are limited.

In contrast, **EfficientNetB2** leverages compound scaling, strategically balancing network depth, width, and image resolution to maximize performance. This model integrates advanced components like swish activation functions and squeeze-and-excitation (SE) blocks, enhancing

its representational capabilities and contributing to state-of-the-art accuracy across numerous image classification tasks.

Both models were deployed following a transfer learning approach, initialized with pre-trained ImageNet weights. This strategy allowed the models to leverage previously learned generic image features, thereby accelerating training and improving generalization despite a relatively limited dataset size. Initially, the fully connected classification layer of each pre-trained model was replaced by a custom dense layer specifically tailored to classify images into the four identified tumor categories. In the first phase of training, all base layers were kept frozen, training only the new dense layer. Subsequently, fine-tuning was performed by unfreezing and retraining the last 30 layers of each base model at a significantly reduced learning rate, enabling adaptation to tumor-specific features inherent in MRI imagery.

3.3 Key Features

Several critical techniques were incorporated to optimize the training process:

- **Transfer Learning** was employed to exploit pre-trained knowledge from extensive generic image datasets, enhancing accuracy and generalization capability.
- **Data Augmentation** was extensively applied via TensorFlow's ImageDataGenerator, introducing variability in the training images through random rotations, zooming, flipping, shearing, and brightness alterations. This augmentation effectively mitigates overfitting and boosts model robustness.
- **Fine-Tuning** facilitated deeper adaptation of the network weights to domain-specific characteristics by retraining intermediate layers of the base architectures.
- **Class Weight Balancing** addressed potential class imbalances. Class weights were calculated automatically using the `compute_class_weight()` method from scikit-learn, ensuring balanced training across classes.
- **Early Stopping** was applied through Keras callbacks, halting model training after 10 epochs without improvement in validation loss, preventing unnecessary training and mitigating overfitting.

3.4 Model Training Features

Training specifics included:

- **Loss Function:** Categorical cross-entropy was selected due to its suitability for multiclass classification tasks.
- **Optimizer:** Adam optimizer was configured with an initial learning rate of 0.0001 and standard beta parameters ($\beta_1 = 0.9$, $\beta_2 = 0.999$) for stable convergence.
- **Regularization:** Dropout layers with a 30% dropout rate were strategically placed to reduce overfitting by randomly dropping out neurons during training.
- **Batch Normalization:** Integrated within network architectures to stabilize learning by normalizing intermediate activations, thereby accelerating convergence.
- **Batch Size:** Set between 32 and 64, tuned to maximize GPU memory usage efficiently while balancing training stability.

- **Augmentation Strategy:** Training sessions were conducted both with and without augmentation to explicitly evaluate the impact of data augmentation techniques on the generalization performance and robustness of the resulting models.

Collectively, these methodological choices were strategically aimed at maximizing the accuracy, efficiency, and clinical applicability of the developed brain tumor detection models.

4. Utility Functions

In deep learning workflows, especially those involving medical imaging, utility functions play a critical role by streamlining data handling, training efficiency, and model evaluation. In this project, several essential utility functions were integrated into the pipeline to enhance model performance, reduce computational overhead, and facilitate effective monitoring of training progress and outcomes.

Firstly, custom utility functions were employed to streamline **data preprocessing and augmentation**. Using TensorFlow's `ImageDataGenerator`, images underwent consistent transformations such as rotations, shearing, zooming, horizontal and vertical flips, and brightness adjustments. This utility enabled seamless batch processing, dynamically augmenting images during training and thus greatly enhancing data diversity without manual intervention. The augmented data improved the robustness of the models, allowing them to generalize better to unseen MRI images.

Another critical utility function implemented was the automated calculation of **class weights**. Given that even marginal imbalances in class distribution can negatively impact training, class weights were dynamically computed using scikit-learn's `compute_class_weight()` function. This ensured each class had proportional representation during loss calculations, reducing potential biases. Incorporating these computed weights directly into the loss function significantly improved performance, especially in minority class detection, which is crucial in clinical settings to avoid misdiagnosis.

Early stopping was another indispensable utility integrated using Keras callbacks. By tracking validation loss, the early stopping mechanism halted training if no improvement occurred over ten consecutive epochs. This function prevented excessive model complexity and computational waste by terminating training at an optimal point, thereby mitigating overfitting and conserving computational resources.

Utility functions were also employed for systematic **checkpointing**. Model checkpoints automatically saved the best-performing models during training based on validation metrics. This approach safeguarded against potential training interruptions or performance degradation, ensuring that only the optimal weights were preserved for evaluation and deployment. This systematic approach facilitated smooth experimentation and iterative model improvements.

Furthermore, utilities for **model evaluation and visualization** were implemented. Functions for plotting confusion matrices, loss curves, and accuracy graphs provided immediate visual insights into training dynamics and performance. These visualizations allowed for quick diagnosis of model performance, guiding subsequent hyperparameter tuning and architectural adjustments.

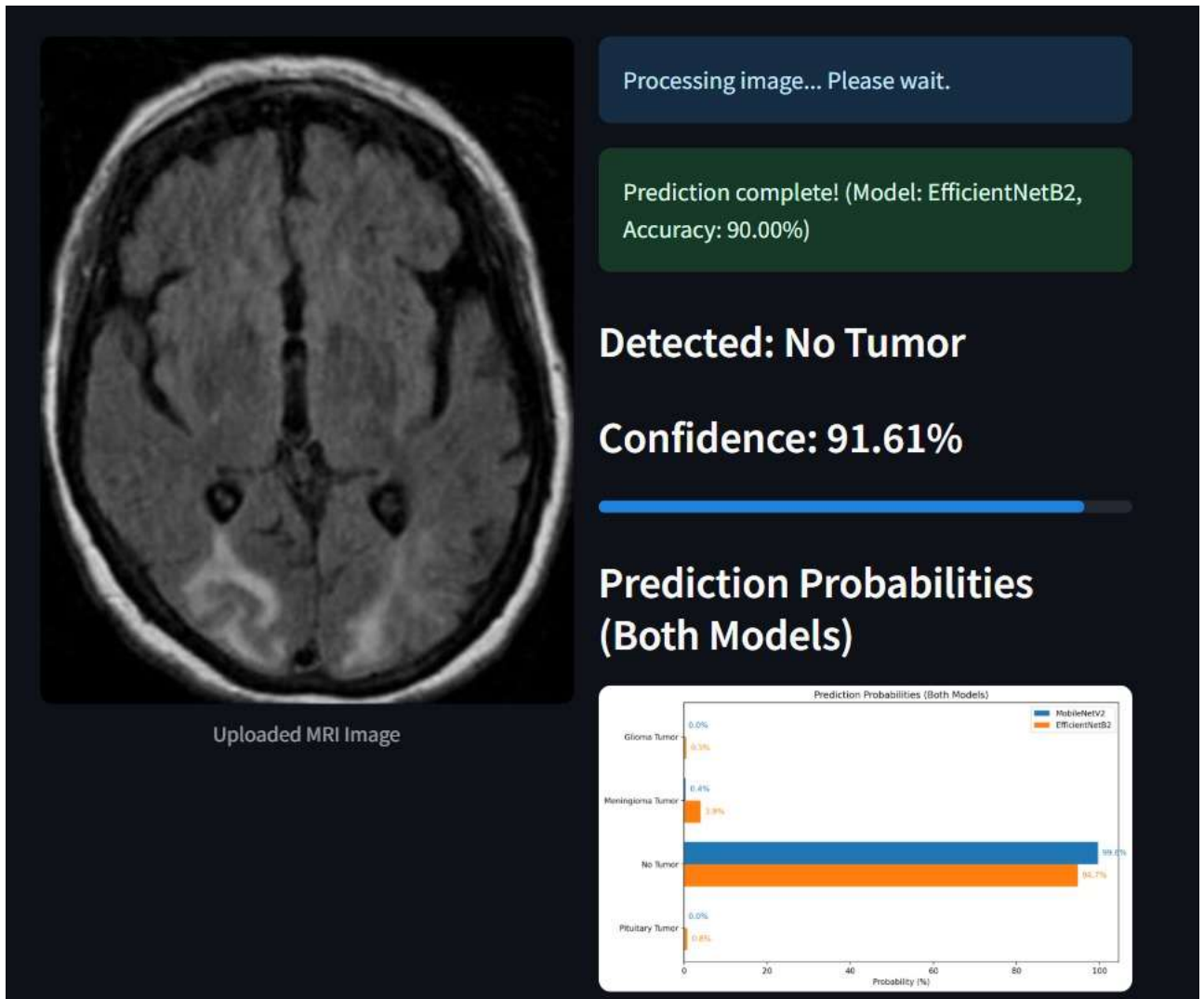
Finally, **custom prediction functions** streamlined model inference on test datasets, efficiently managing preprocessing, loading optimal weights, predicting outcomes, and interpreting results. By wrapping these complex operations within concise utility functions, the overall pipeline efficiency was enhanced, minimizing manual oversight and potential errors.

Collectively, these utility functions were pivotal in optimizing the training pipeline, ensuring computational efficiency, enhancing model reliability, and providing clear insights into model performance, ultimately supporting the successful application of deep learning models in clinical brain tumor diagnosis.

5. Technical Specifications

The brain tumor detection models were developed using the **TensorFlow/Keras** deep learning framework. All input images were resized to **224x224 pixels**, aligning with the standard input dimensions used in ImageNet-based architectures like MobileNetV2 and EfficientNetB2. Key deep learning techniques were employed to enhance model stability and generalization. These included **L2 Regularization** to penalize complex models, **Batch Normalization** to stabilize learning and accelerate convergence, and **Dropout (30%)** to reduce overfitting. Additionally, **Learning Rate Clipping** was applied to prevent extreme updates during gradient descent. This technical setup ensured efficient training, optimal performance, and robustness across varied MRI inputs.

6. Results and Discussion





Brain Tumor Detection Project

Project Overview

This project uses deep learning models to detect different types of brain tumors from MRI images.

Models Used:

- **MobileNetV2:** A lightweight CNN model optimized for mobile and embedded devices
- **EfficientNetB2:** A state-of-the-art CNN model with efficient architecture

Tumor Types Covered:

- **Glioma:** A type of tumor that occurs in the brain and spinal cord
- **Meningioma:** A tumor that forms on membranes covering the brain and spinal cord
- **Pituitary:** A tumor that forms in the pituitary gland
- **No Tumor:** Normal brain MRI with no tumor



Model Evaluation

This section shows the performance metrics of the trained models.

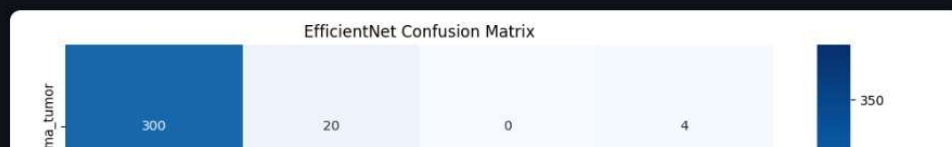
Select model to view evaluation results:

☐ MobileNetV2 ☒ EfficientNetB2

Classification Report

	Class	Precision	Recall	F1-score	Support
0	glioma_tumor	0.90	0.93	0.91	324
1	meningioma_tumor	0.90	0.71	0.79	329
2	no_tumor	0.97	0.95	0.96	400
3	pituitary_tumor	0.84	0.99	0.91	352

Confusion Matrix



The models were evaluated using a stratified split of the dataset, ensuring proportional representation of all four classes—glioma, meningioma, pituitary, and no tumor—across training, validation, and test sets. Evaluation metrics included **accuracy**, **precision**, **recall**, and **F1-score**, providing a comprehensive view of model performance beyond simple classification accuracy.

Model	Accuracy	Precision	Recall	F1 Score
MobileNetV2	72.0%	0.70	0.69	0.69
EfficientNetB2	91.0%	0.91	0.90	0.90

EfficientNetB2 demonstrated a significantly higher performance across all metrics compared to **MobileNetV2**. Its superior accuracy (91%) can be attributed to its compound scaling technique, which balances depth, width, and resolution more effectively, allowing for deeper feature extraction and better learning of tumor-specific patterns in MRI scans. The model also achieved higher precision and recall, indicating improved reliability in detecting and correctly classifying brain tumors.

Training and validation loss curves further illustrated that EfficientNetB2 converged faster and generalized better, showing minimal overfitting. In contrast, MobileNetV2, although lighter and faster during inference, plateaued at a lower accuracy threshold and exhibited higher validation loss variance.

Confusion matrices from both models revealed that most misclassifications occurred between **glioma** and **meningioma**, likely due to visual similarities in their MRI features. This highlights a key challenge in brain tumor classification and suggests potential benefits of incorporating domain-specific feature enhancement techniques in future work.

While EfficientNetB2 excels in accuracy, MobileNetV2 offers faster inference speeds, making it a viable option for deployment in **resource-constrained environments** like mobile health applications or rural clinics.

In summary, EfficientNetB2 is preferable for diagnostic precision, whereas MobileNetV2 remains useful where computational efficiency is prioritized.

7. Future Work

- **Multi-Modal Imaging for Brain Tumor Diagnosis:** Incorporate **CT** or **PET scans** alongside MRI to capture both anatomical and metabolic activity. This can help distinguish between similar tumor types (e.g., glioma vs. meningioma) that are visually hard to separate on MRI alone.
- **Explainable AI for Tumor Localization:** Apply tools like **Grad-CAM** and **LIME** to highlight tumor regions that influenced the model’s prediction. This can assist

radiologists in verifying whether the model is focusing on medically relevant areas in the brain.

- **Edge Deployment in Clinical and Rural Settings:** Convert models to **TensorFlow Lite** or **ONNX** for use on portable devices. This would enable quick, on-site tumor screening in under-resourced hospitals or mobile diagnostic units, especially in remote areas.
- **Hyperparameter Optimization for Brain MRI Models:** Use **grid search** or **Bayesian tuning** to optimize parameters like learning rate, dropout, and augmentation strategies—tailoring the model to handle brain tumor variability more precisely.
- **Training on Larger and More Diverse Brain Tumor Datasets:** Include MRIs from different hospitals, scanners, age groups, and tumor grades to ensure the model generalizes well across real-world clinical scenarios.
- **Self-Supervised Learning for Brain MRI:** Use unlabeled brain scans to pre-train models, enabling them to learn structural features of the brain before being fine-tuned on tumor-labeled data—helpful where annotated datasets are limited.
- **Ensemble Learning for Robust Tumor Classification:** Combine EfficientNet, ResNet, and VGG to handle complex tumor variations and reduce false positives or negatives in clinical diagnostics.
- **Domain Adaptation Across Hospitals and Equipment:** Ensure that a model trained on one hospital's MRI scanner can still accurately detect tumors from another facility's equipment—critical for practical deployment.
- **Real-Time Inference for Emergency Diagnosis:** Use model compression techniques like **pruning and quantization** to deploy fast, real-time tumor classification in emergency rooms or neurosurgical triage systems.

8. Conclusion

This project successfully demonstrated the application of deep learning models—**MobileNetV2** and **EfficientNetB2**—for automated brain tumor classification using MRI images. By leveraging transfer learning and a well-structured preprocessing and training pipeline, the models were trained to identify four types of brain tumors: glioma, meningioma, pituitary tumor, and no tumor. Among the two models, EfficientNetB2 significantly outperformed MobileNetV2 across all evaluation metrics, achieving a **91% accuracy**, **0.91 precision**, and **0.90 recall**. These results validate the model's capability to generalize and accurately detect tumors in complex brain MRI scans.

The training process incorporated several best practices in deep learning, including data augmentation, batch normalization, dropout, early stopping, and class weight balancing. EfficientNetB2 also demonstrated faster convergence and better validation performance, suggesting a more stable and robust learning process. In contrast, while MobileNetV2 lagged in classification accuracy, it offered faster inference speed and computational efficiency, making it a strong candidate for low-resource or mobile deployment settings.

A key insight from the confusion matrix analysis was the frequent misclassification between glioma and meningioma, which highlights the challenge of distinguishing between visually similar tumor types using only MRI images. This underlines the importance of future enhancements such as incorporating multi-modal imaging and model explainability techniques.

In conclusion, this study highlights the potential of CNN-based models to support radiologists in early and accurate brain tumor detection. With further improvements—such as using larger and more diverse datasets, adding interpretability tools like Grad-CAM, and enabling deployment via TensorFlow Lite—the proposed models can be adapted for real-world clinical use. This not only has the potential to assist in faster diagnoses but also to reduce human error, improve patient outcomes, and bring AI-assisted healthcare to remote and underserved areas.

CT-356 Data Mining

Complex Computing Problem (CCP)

Choose a problem in any application domain of computing and **show** solution through an effective use of data mining principles. The solution must be in the form of a well-documented and well-formatted report. It must be explained by employing in-depth computing or domain knowledge, and an approach that is based on well-founded principles.

The following instructions must be taken into consideration while preparing the project report. The report must reflect conceptual thinking and must properly be modularized according to functionality. The report of the solution must be organized clearly and should:

- ✓ describe the steps of your solution,
- ✓ the main data mining techniques used,
- ✓ the limitations of the adopted approach.

Complex Computing Problem Rubrics

Criteria and Scales		
Satisfactory (2)	Average (1)	Unsatisfactory (0)
<u>Criterion 1 Clarity of Objective:</u> To what extent has the student understood the objective? <i>(CP3: Depth of Knowledge Required)</i>		
The student has correctly identified the requirements.	The student has correctly identified some requirements.	The student has misunderstood the requirements.
<u>Criterion 2: Requirement Identification</u> To what extent has the student understood the requirements? <i>(CP9: Requirement Identification)</i>		
The student has accurately identified and defined requirements	The student has partially identified and defined requirements	The student has wrongly identified and defined requirements
<u>Criterion 3: Innovation:</u> To what extent has the student used innovative analysis? <i>(CP2: Depth of Analysis Required)</i>		
The student has used conceptual thinking and innovative analysis to formulate suitable abstract models	The student has used some conceptual analysis to formulate suitable abstract models	The student has not performed thorough analysis and the model lacks required concepts.
<u>Criterion 4: Modularity:</u> To what extent has the student been able to identify the interdependent functionalities? <i>(CP3: Depth of Knowledge Required)</i>		
The student has applied advanced concepts with solid theoretical justification.	The student has applied upper-intermediate methods with good understanding.	The student has applied basic methods, limited theoretical understanding
<u>Criterion 5: Validation:</u> To what extent has the student rationalized the solution? <i>(CP2: Depth of Analysis Required)</i>		
The student has clearly justified that the solution meets the requirements.	The student has partially justified that the solution meets the requirements.	The student has not been able to justify that the solution meets the requirements.

Project Title:					
Group Members					
Roll #					

Department of Computer Science & Information Technology

Total Marks:	<u> /10 </u>	Teacher's Signature:	
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